

Practical Deep Learning Models for QIM-based VoIP Steganalysis

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ABSTRACT

Quantization index modulation (QIM) based VoIP steganography can conceal secret information in VoIP streams. Malicious users could use this technology to conduct illegal activities, threatening network and public security. Hence, practical steganalysis models that could detect QIM-based VoIP steganography are urged to be developed. In recent years, deep learning (DL) models have been investigated for this task, and exciting outcomes have been achieved. However, existing models are far from practical. Two major challenges are required to be addressed. First, there is still significant room for improvement in detection accuracy. Second, studies that balance the detection accuracy and response time are still insufficient. In this context, our main research topic fits in the QIM-based VoIP steganalysis theme, which aims to detect QIM-based steganography in VoIP streams in a fast and accurate manner.

CCS CONCEPTS

• **Networks** → **Network security**.

KEYWORDS

VoIP, steganography, steganalysis, covert channel, information hiding

ACM Reference Format:

Cheng Zhang. 2024. Practical Deep Learning Models for QIM-based VoIP Steganalysis. In *Proceedings of the 32nd ACM International Conference on Multimedia (MM '24)*, October 28–November 1, 2024, Melbourne, VIC, Australia. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3664647.3685646>

1 INTRODUCTION

Misuse of quantization index modulation (QIM) based VoIP steganography is a potential threat to the network and public security. For example, One could send virus instructions to target systems without being detected through QIM-based VoIP steganography. Law-breakers, such as drug dealers and human traffickers, could establish covert communications using QIM-based VoIP steganography. Compared to common digital carriers, the use of VoIP streams has several benefits. First, the high payload of VoIP streams allows QIM-based VoIP steganography to hide huge amounts of information. Next, in most real-life scenarios, the VoIP stream data packages are discarded

immediately after use, ensuring QIM-based VoIP steganography's confidentiality. Moreover, many applications (such as Skype and WeChat) and devices (such as laptops and cell phones) support VoIP calls, providing QIM-based VoIP steganography with countless application scenarios. In summary, QIM-based VoIP steganography is extremely practical, Highly confidential, and hard to detect/trace.

A classical application scenario for VoIP steganography and steganalysis is shown in Figure 1. In VoIP scenarios, the speeches are compressed (encoded) through speech coders to save bandwidth. QIM-based steganography alters the encoding procedure to hide secret information, which ultimately changes the value of a specific type of encoding codewords - the line spectrum frequency (LSF) codewords. QIM-based VoIP steganalysis is devoted to detecting steganographic VoIP streams in time. During the VoIP call, the communication channel is inspected. Some speech samples are fed into the steganalysis model. Suppose any of these samples are determined as stego (steganographic), which means they are modified by steganography. In that case, the result will be sent to an administrator to support the decision making procedure (e.g., cut the communication channels).

Despite the subtle changes caused by steganography, they disrupt the correlations among the codewords. Thus, the key factor for QIM-based VoIP steganalysis is to capture the correlation changes among the codewords. Traditionally, handcrafted features [2, 3] (such as frequency and distribution) are designed and extracted from speech samples for training classifiers like support vector machines. However, the handcrafted features are hard to design and can hardly describe the codeword correlations in a comprehensive manner. Therefore, researchers attempted deep learning (DL) models in QIM-based VoIP steganalysis and achieved exciting results [5]. Since then, instead of using handcrafted features, most of the studies directly feed the LSF codewords into carefully designed deep learning models to automatically extract and learn the correlation features for steganalysis. Even though the existing deep learning models have greatly improved the detection performance compared to the handcrafted features, they are far from practical in terms of detection accuracy, model size (scalability), and response time.

To address these issues, our research revolves around two themes. First, investigate various DL architectures based on the characteristics of VoIP speech encoding and explore the possible connections between DL architectures and encoding characteristics. Previous studies have focused more on the application of advanced DL models in QIM-based VoIP steganalysis but have overlooked the characteristics of the encoding procedure and the rationale behind the performance. We believe that by considering the characteristics of the encoding procedure, more reasonable DL architectures and features could be designed. Meanwhile, analyzing the rationale behind

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MM '24, October 28–November 1, 2024, Melbourne, VIC, Australia

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ACM ISBN 979-8-4007-0686-8/24/10

<https://doi.org/10.1145/3664647.3685646>

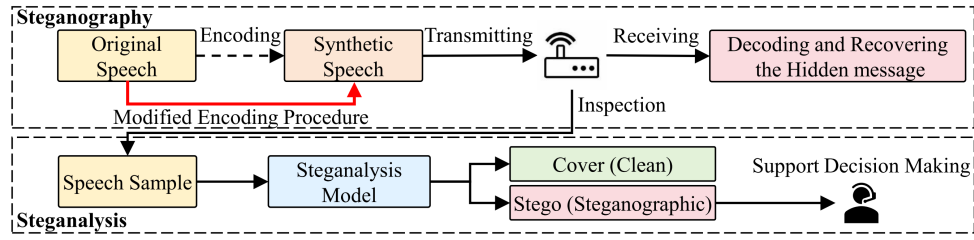


Figure 1: VoIP steganography and steganalysis. VoIP steganography algorithms modify the original encoding procedure to hide the secret message in the encoding parameters. The synthetic speech is then transmitted over the Internet. Once received, the receiver decodes the synthetic speech to obtain the encoding parameters and recover the hidden message. In VoIP steganalysis, speech samples are collected from an inspected communication channel. These samples are then fed into a steganalysis model for testing. Stego test results will be immediately sent to the authorities to support the decision making process.

the performance (possible connections between DL architectures and encoding characteristics) could potentially guide future studies. Second, with the knowledge learned in the first research theme, which guarantees detection accuracy, we establish further studies on limiting the model size and the response time.

2 PROBLEM DESCRIPTION

To perform VoIP steganalysis, the parameters (codewords) are extracted following the decoding procedure. Each speech sample is stored as a matrix. The rows of the matrix are the frames. The longer the speech sample is, the more rows (frames) it contains. The columns of the matrix are the codewords. The frames contain the same number of codewords.

Generally, the adoption of QIM-based VoIP steganography only makes slight changes to the LSF codewords. The key idea of QIM-based VoIP steganalysis is to capture the correlation changes caused by these slight changes. In theory, longer speech samples contain more codeword correlations than shorter speech samples. Consequently, detecting longer speech samples is much easier. However, in real-life scenarios, steganalysis models are required to detect speech samples as short as possible to assure timeliness. Currently, detecting QIM-based VoIP steganography in short speech samples in a fast and accurate manner is still a challenging and unsolved research question.

3 RELATED WORK

3.1 Detection Accuracy

In some early studies, researchers used handcrafted features to model the statistical characteristics of the encoding codewords, such as the codeword distribution [2, 3]. [5] first introduced deep learning models into the QIM-based VoIP steganalysis task and achieved outstanding outcomes. Inspired by their works, many novel and more complicated network architectures have been widely investigated to improve detection accuracy. For example, [10] proposed a hierarchical network architecture to extract multi-scale codeword correlation features. [13] proposed a dual-stream network architecture to extract multi-domain features. [4] combined the long short-term memory (LSTM) network and the convolutional neural network (CNN) to capture the contextual information and the local

characteristics of the codewords. [8] introduced a dual-thread architecture to obtain the multi-dimensional codeword correlation features.

In summary, these studies adopted sophisticated deep learning models to capture the in-frame and cross-frame codeword correlations (among the LSF codewords) that were modified by the QIM-based VoIP steganography as the critical clue for steganalysis. Despite the accuracy being improved, two major problems are still unsolved. First, the existing studies all take the LSF codewords as the input. However, the LSF codewords are not independently used during the VoIP encoding and decoding procedure. Certain correlations existed among the LSF and other codewords, which could potentially be used for steganalysis but have never been explored by previous studies. Second, resource consumption has been ignored in many related studies. With the network architectures becoming deeper and wider, some models are unpractical in real-life scenarios.

3.2 Resource Consumption

Resource consumption is an important factor that would impact the models' practicability since many devices that support VoIP calls are mobile devices with limited computation resources. Currently, studies that investigate lightweight architectures are rare. [12] proposed a lightweight model that only consists of codeword embedding and multi-head attention mechanisms. Despite the model's response time being fast, the detection accuracy is harmed as the sequential information (cross-frame correlations) could not be captured by the proposed model. [11] first introduced the knowledge distillation concept to VoIP steganalysis and proposed a lightweight student network. Similar to the previous study [12], the student network only focuses on the in-frame correlations.

In summary, the studies on lightweight QIM-based VoIP steganalysis are far from enough. To limit resource consumption, existing studies ignore the sequential information contained in VoIP streams. Lacking sequential information harms the detection accuracy of the proposed models. In real-life scenarios, the number of VoIP calls is countless. Even a small drop in the detection accuracy would result in a huge amount of false alarms (or missed detections). Moreover, many advanced lightweight design concepts and architectures such as MobileViT [6] have never been considered in the QIM-based VoIP Steganalysis domain.

4 RESEARCH QUESTIONS

This section introduces two major research questions being investigated and their potential impact on QIM-based VoIP steganalysis.

RQ1: How can we leverage the correlations between the LSF and other codewords to enhance the detection accuracy?

The LSF codewords are not independently used during the VoIP encoding and decoding procedure. Existing studies all take LSF codewords as input, ignoring the correlations among LSF and other codewords. Intuitively speaking, the detection accuracy could potentially be further improved if these correlations could be captured. However, the other codewords also contain many redundant features that are not related to steganography and do not contribute to steganalysis. Investigating methods to utilize the useful correlation features among the LSF and other codewords while suppressing the noise contained in the other codewords would help improve the detection accuracy and guide future research in this domain.

RQ2: How can the detection accuracy and resource consumption be balanced to make the steganalysis models more practical?

Lightweight models have not been sufficiently explored in the QIM-based VoIP steganalysis domain. Most existing studies focus on detection accuracy and ignore resource consumption, which affects the models' practicability. Some lightweight studies fail to capture the sequential features, harming detection accuracy. Meanwhile, many advanced lightweight concepts and architectures in the deep learning domain have never been considered in the QIM-based VoIP steganalysis task. Limiting resource consumption while maintaining high detection accuracy could make the models more practical and expand their application scenarios.

5 PROPOSED APPROACHES, EXPERIMENTS, AND RESULTS

Dataset: Most studies use the VoIP speech dataset published in [5]. However, this dataset only contains clean speech samples. To address this issue, we released a standard dataset that contains both cover and stego speeches in our previous work [8], which could facilitate fair comparisons of steganalysis methods in future studies. Moreover, it would streamline the experimental process for researchers. The dataset comprises 171,204 Chinese and 306,248 English speech samples (10-second, stored as matrices). The speech samples are encoded with the widely adopted speech coder G.729 [7]. Each row of the matrix (one speech sample) is one 10ms frame that contains 15 encoding codewords. Hence, a 10-second speech sample can be represented as a matrix of 1000 rows and 15 columns. Both the Chinese and English datasets consist of 11 sub-datasets, including a cover (clean samples) dataset and 10 stego (steganographic samples) datasets. The stego datasets are created through a benchmark QIM-based VoIP steganography named CNV [9] (derived from [1]) that has been used in almost all the studies in this domain (still challenging for recently proposed models). The 10 stego datasets are created with different embedding rates ranging from 10%, 20%, to 100%. To ensure confidentiality, CNV only embeds 1 bit per frame. Hence, the embedding rate is defined as the number of embedded frames to the total number of frames. For example, a 50% embedding rate 10-second stego speech sample contains 1000 frames, in which 500 frames are embedded with secret bits.

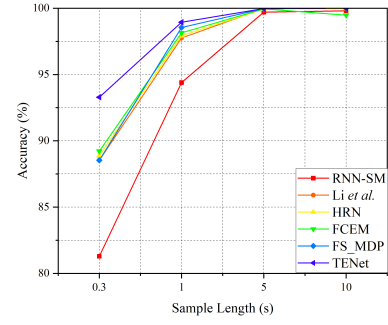


Figure 2: The experimental results of 50% embedding rate English speeches under different sample lengths.

RQ1: How can we leverage the correlations between the LSF and other codewords to enhance the detection accuracy?

We proposed a framework named **TENet** [15] (Transformer Encoder based Network) to leverage the correlations between the LSF and other codewords. **Rationale:** We observe that the LSF codewords are not independently calculated during the encoding procedure [7]. Certain correlations could exist between the LSF codewords and others. Capturing these correlations might help improve the detection accuracy. **Problem encountered:** Noting that CNV only changes the LSF codewords by 1 bit per frame (extremely slight). Although correlations among the LSF codewords and others might help improve the detection accuracy, the use of extra codewords introduces extra noises contained in them. The research problem we faced was extracting useful information for steganalysis while excluding the noises. **Solution:** First, We leverage the transformer encoder in the beginning for the following reasons: (1) The speech data are sequential, the transformer exhibits excellent performance on sequential data and has never been explored in the QIM-based VoIP task. (2) As the input speech sample is not always 10 seconds, the transformer has the advantage that it can process input with different lengths easily. (3) The multi-head attention in the transformer allows it to capture long-term correlations, which is suitable for our task. In our case, the frames that contain codewords are sequential and correlated (long-term correlations). (4) As our task is a binary inference task instead of a generative task, we exclude the transformer decoder. We take the hidden features (codeword correlations) encoded with the transformer encoder as the detection clue for steganalysis. Then, we establish preliminary experiments by directly feeding all the encoding codewords, which generates terrible results and proves that using extra codewords would introduce extra noise. To suppress the noise, we hypothesize that categorizing the codewords in the data pre-processing procedure might help the model find the boundaries and differences among the codewords instead of treating them equally. In order to categorize the codewords, we combined the encoding procedure [7]. Following the encoding procedure, we categorize the codewords by their usage: The LSF (signal synthesis), pitch delay (pitch analysis), code index (signal excitation), and gains codewords (scaler of code index). We carefully designed the architecture of TENet based on these four types of codewords and achieved outstanding performance [15]. RNN-SM [5], HRN [10], FCEM [12], FS_MDP

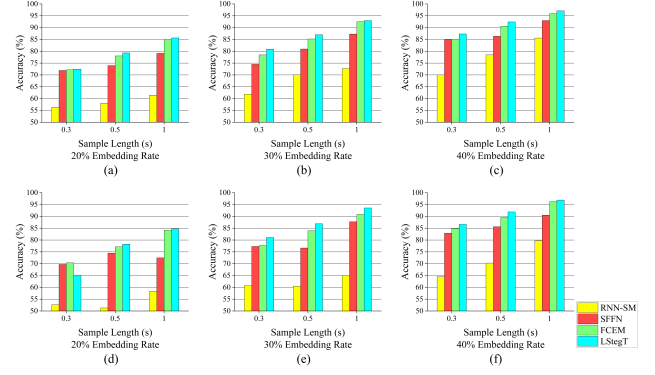
Table 1: Detection Performance of Lightweight Models with 0.3-second English Speech Samples under 40% Embedding Rates

Embedding Rate	Method	Accuracy (%)	F1 Score (%)	FP Rate (%)	Response Time (ms)
40%	RNN-SM	70.01	69.43	28.11	0.1934
	SFFN	84.88	84.69	13.92	0.2085
	FCEM	84.98	84.62	12.68	0.1911
	LStegT	87.31	87.28	12.44	0.1758

[8], and the method proposed by Li *et al.* [4] are implemented and compared with TENet. We demonstrate part of the experimental results in Figure 2 (full experimental results can be found in [15]). **RQ2: How can the detection accuracy and resource consumption be balanced to make the steganalysis models more practical?**

We proposed a framework named **LStegT** (Lightweight Steganalysis Transformer), whose design concept is inspired by MobileViT. **Rationale:** The success of MobileViT proves that separating the extraction procedure of different features and properly designing the corresponding blocks could help reduce the number of learning parameters required by the deep learning model. Also, the combination of CNNs and ViTs (vision transformers) could build a lightweight and low-latency network. **Problem encountered:** The MobileViT is not suitable for QIM-based VoIP steganalysis for the following reasons. First, the data sizes in vision tasks are much larger than the VoIP speeches. For example, a 4k RGB image is a $3 \times 4,096 \times 2,160$ matrix, while a 10-second VoIP speech sample (encoded with G.729) is a $1,000 \times 15$ matrix only. Moreover, the QIM-based VoIP steganography task is binary (either cover or stego), which is much easier than many vision tasks. Intuitively, MobileViT contains more learning parameters than necessary for the QIM-based VoIP steganalysis task. **Solutions:** We hypothesize that the correlation features used in QIM-based VoIP steganalysis could be categorized into two parts: (1) The codeword correlations within the same frame (local correlations) and (2) the correlations across different frames (global correlations) that contains sequential information. Based on this hypothesis, we separately designed two correlation extraction blocks. First, the local correlation extraction block utilizes the codeword embedding and deep-wise separate convolutions to capture the in-frame codeword correlations. Second, a transformer encoder-based block is designed to capture the sequential information (cross-frame correlations). Since each of these blocks is only responsible for capturing their corresponding features, the number of learning parameters required within these blocks could be further reduced.

We demonstrate part of our experimental results in Table 1 and Figure 3. As shown in Table 1 and Figure 3, LStegT outperforms some relatively lightweight models (including RNN-SM [5], SFFN [13], and FCEM [12]) in most experimental scenarios with an extremely fast response time. Worth noting that in Figure 3(d), LStegT has lower detection accuracy than FCEM when the embedding rate is 20%. One possible reason is the local and global correlations are extracted separately. When the embedding rate is extremely low, it is hard for the local correlation extraction block to distinguish the valuable and redundant features without the cross-frame information as references. Also, we observe that LStegT only slightly

**Figure 3: Detection performance of different methods under multiple embedding rates and sample lengths. English: (a), (b), (c). Chinese: (d), (e), (f).**

improves the performance when the sample length is 1-second. Due to the minimal number of training parameters, longer speech samples contain more useful features than LStegT could capture. Therefore, increasing the sample length to 1-second does not significantly improve LStegT's performance. Full experimental results can be found in [14].

5.1 Future Work

(1) With the existing findings, it is worth exploring a key factor-led (LSF codewords-led) architecture that not only knows the difference between different types of codewords but also learns their importance level, which might further improve the detection accuracy [Under submission]. (2) Current models are still considered as resource-costly for some small devices. I am working on formulating a lightweight, fast, and accurate architecture (purely consisting of convolutional neural networks) that could achieve superb performance with a limited model size [Under submission]. (3) Currently, we found the training procedure in our experiments is costly and takes months. We would like to combine technologies in the transfer learning and pre-training domain. First, we treat each embedding rate as individual sub-domains. Then, we plan to use adequate data (clean and mixed embedding rate) to pre-train a preliminary model. Finally, for each of the sub-domains (embedding rates), we only take a small amount of data to fine-tune the model. If successful, this could save a huge amount of training resources in future studies [Ongoing]. (4) We observe that some mobile devices could hardly execute LSTM or transformer-based networks. These architectures are necessary for the sequential information (cross-frame correlations) to be fully extracted, which harms the detection accuracy. Based on this observation, we plan to transfer VoIP speech samples into images and encode the sequential information into the images' channels (such as RGB). We hope to expand the receptive field of CNNs through this transformation [Ongoing].

ACKNOWLEDGMENTS

This work was supported in part by the National Natural Science Foundation of China under grant No. 61673384.

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